NIPB NEWSLETTER

ICAR-National Institute for Plant Biotechnology,
Pusa Campus, New Delhi-110012

Vol. 06 | No. 1-2 2023

FROM DIRECTOR'S DESK

Plant biotechnology has witnessed phenomenal development in the recent decades. In agriculture, application of biotechnological tools and techniques have been more universal across the disciplines. ICAR- National Institute for Plant Biotechnology (NIPB) has been the centre stage of agricultural biotechnology since its inception in 1985. It has played a pivotal role in adopting the biotechnological developments in Indian agriculture. It has undertaken basic scientific research, devised tools and techniques, generated products and more importantly generated the trained human resource for undertaking plant biotechnology research across the research institutes in the country.

ICAR-NIPB has been mandated to undertake research in five major crops rice, wheat, mustard, pigeonpea and chickpea in addition to extending hand holding support to other crop-based institutes of ICAR and SAUs. This centre has been the pioneer in crop genome sequencing and completed genome sequences of 9 crops either through international consortium or by its indigenous efforts. Utilizing the vast resources of genome sequences and genomic tools, it has characterized and validated a large number of QTLs, genes and promoters governing diverse agronomically important traits. Discovery of QTLs for high grain number and anaerobic germination in rice, determinacy and early flowering in pigeonpea, drought tolerance in chickpea are some of the recent and potential research developments. Continuing with its legacy in playing the lead role in coordinating transgenic development in crop plants undertaken by 15 institutes of NARES, the institute has recently completed the event selection trial of pod borer resistant pigeonpea.

Wild crop species have been the rich reservoir of many agronomically important genes which are particularly important in engineering climate resilience in crop species. Over the years this institute has created a large reservoir of more than 1200 Crop Wild Relatives (CWRs) in rice and more than 25 in mustard. In addition, the CWRs are being studied through genomics and metabolomics which has led to discovery of novel genes like, IGMT (indole glucosinolate methyl transferase) from *Diplotaxis erucoides* and metabolites like epicatechin-3-gallate from *Cajanus platycarpus* which has conferred tolerance to Alternaria blight in mustard and pod borer resistance in pigeonpea, respectively.

Improving precision in introgression of trait and reducing the breeding cycles in variety development through Marker Assisted Selection (MAS) is another potential application of biotechnology in agriculture. This institute is known for its seminal and frontline expertise in developing and using many of the important markers and SNP genotyping gene chips in rice, pigeonpea, mango etc.

It is my proud privilege to release this newsletter to the readers. This newsletter presents the window to look into the diverse activities and developments in this institute during the period January-December 2023. I congratulate the Chief editor and the entire editorial team who have contributed in meticulous composition of this issue. I place on record my sincere gratefulness to Dr. Himanshu Pathak, Secretary, DARE and DG, ICAR; Dr. T.R. Sharma, DDG (CS) and Dr. D.K. Yadava, ADG (Seeds) for their continued support and guidance to this institute. I also congratulate all the staff of NIPB including scientific, technical, administrative and supporting staff whose team work has led to several significant achievements of this institute. I hope that this issue of the Newsletter would be received well by readers across the domains. Any critical comments and/or suggestions would be appreciated.

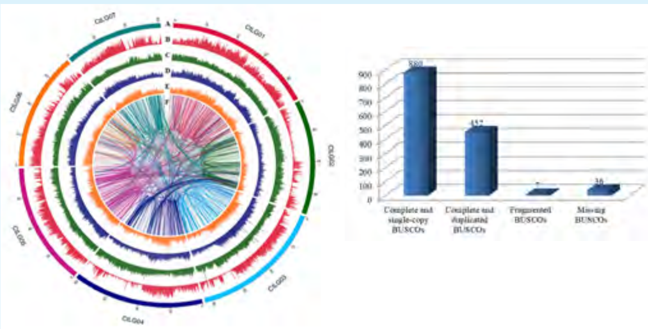
(Ramcharan Bhattacharya)

CONTENTS

The chromosome-scale genome assembly of cluster bean provides molecular insight into edible gum (galactomannan) biosynthesis family genes

ICAR-NIPB scientists have sequenced and assembled a very high quality genome of guar (*Cyamopsis tetragonoloba* L. Taub), a vegetable and fodder crop. A popular cluster bean cultivar, RGC-936, was sequenced by Illumina, 10X Genomics, and Oxford Nanopore technologies and a final assembly of 550.31 Mbp (94% of the estimated genome size of ~ 580 Mb estimated through flow cytometry) organized into 58 scaffolds (7 chromosomes and 51 unplaced scaffolds) was made. Further comparative transcriptomics analyses revealed pod-specific up-regulation of genes encoding enzymes involved in galactomannan biosynthesis. The high-quality chromosome-scale cluster bean genome assembly will facilitate understanding of the molecular basis of galactomannan biosynthesis and aid in genomics-assisted improvement of cluster bean.

Scientific Reports 13: 9941 (2023)

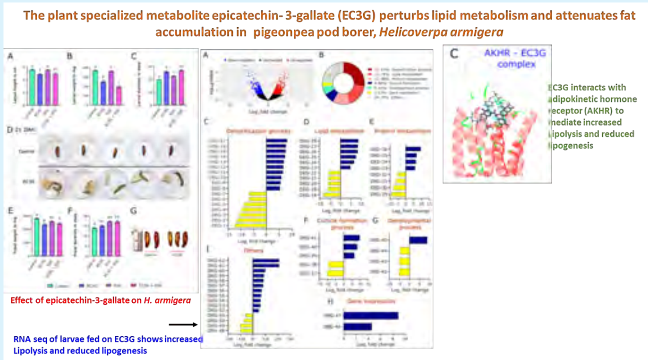


The cluster bean genome features

The plant specialized metabolite epicatechin-3-gallate (EC3G) perturbs lipid metabolism and attenuates fat accumulation in pigeonpea pod borer, *Helicoverpa armigera*

Overproduction of catechins in the pod borer-resistant pigeonpea wild relative, *Cajanus platycarpus* during continued herbivory prodded us to assess their underlying molecular effect on *H. armigera*. Significant reduction in the larval and pupal growth parameters was observed when reared on artificial diet incorporated with 100 ppm EC3G *vis a vis* 100 ppm EGC and EGC+EC3G. Comparative RNAseq analyses of larvae that fed on normal and EC3G-incorporated diet showed possible effect of EC3G on fat metabolism. Differential accumulation of stearic and oleic acid in EC3G-fed and control larvae/adults ascertained perturbation in lipogenesis. Supported by modelling, molecular docking and simulations, we demonstrate the possible involvement of the insect adipokinetic hormone receptor in the EC3G-mediated response.

International Journal of Biological Macromolecules 231: 123325 (2023)



Role of epicatechin-3-gallate (EC3G) in the management of pigeonpea pod borer

^{15}N influx-based identification and functional characterization of a high-affinity nitrate transporter gene, *TaNRT2.1-6B*, of bread wheat

Genome-wide identification of NRT2 family genes and gene expression studies of different members of TaNRT2 family genes led to the identification of a high-affinity nitrate transporter gene (*TaNRT2.1-6B*) expressing significantly high in root tissues. The high-affinity nitrate transport system has been found to contribute significantly at major physiological growth stages of wheat both at optimum (ON) and limiting (LN) conditions in wheat, based on precise phenotyping for nitrate uptake using ^{15}N-labelled N source. *TaNRT2.1-6* was functionally validated employing complementation approach using a corresponding T-DNA Arabidopsis mutant (*atnrt2.1*). Recovery of ^{15}N influx (~ 1.5x) in the complemented line of the mutant validated the findings.

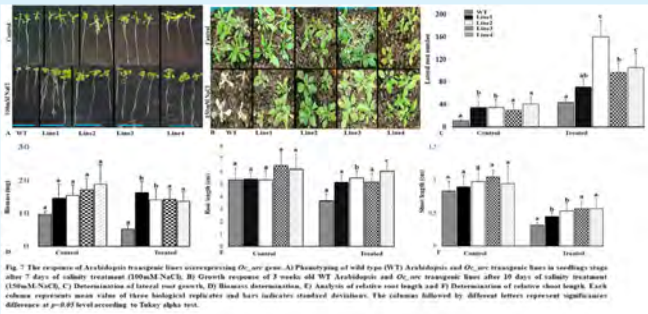
Environmental and Experimental Botany 207: 105205 (2023)

Identification and functional validation of TaNRT2.1-6B gene of bread wheat

Allantoin improves salinity tolerance in Arabidopsis and rice through synergistic activation of abscisic acid and brassinosteroid biosynthesis

We demonstrated the roles of exogenous application of allantoin and increased concentration of endogenous allantoin in rendering salinity tolerance in rice and *Arabidopsis* respectively, via., induction of abscisic acid (ABA) and brassinosteroid (BR) biosynthesis pathways. Exogenous application of allantoin (10 µM) resulted in salt-tolerance (100mM-NaCl) of salt-sensitive rice genotype (IR-29). Transcriptomic data after exogenous supplementation of allantoin under salinity stress showed induction of ABA (*OsNCED1*) and BR (*Oscytochrome P450*) biosynthesis genes in IR-29. Further, the key gene of allantoin biosynthesis pathway, *urate oxidase* of the halophytic species, *Oryza coarctata* was found to induce ABA and BR biosynthesis genes when over-expressed in transgenic *Arabidopsis*.

Plant Mol. Biol. 112(3): 143-160 (2023)

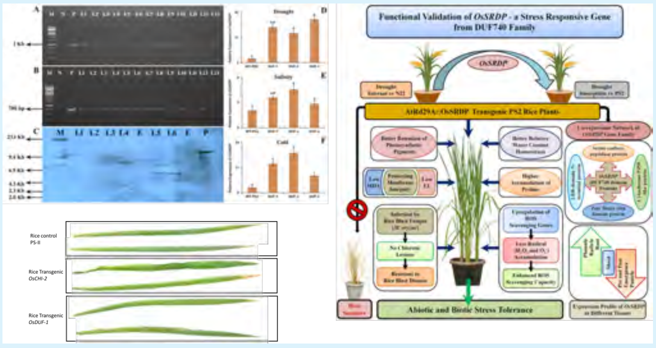


The response of Arabidopsis transgenic lines over expressing *Oc_urc* gene

Overexpression of a DUF740 family gene (LOC_Os04g59420) imparts enhanced climate resilience through multiple stress tolerance in rice

ICAR-NIPB scientists investigated the function of a novel gene LOC_Os04g59420, be longing to DUF740 family (*OsSRDP-Oryza sativa Stress Responsive DUF740 Protein*) from rice. Transgenic plants of the rice *OsSRDP* gene, driven by a stress-inducible promoter AtRd29A, were developed in the background of cv. Pusa Sugandh 2 (PS2) and tested for multiple stress tolerance. The homozygous transgenic plants showed better resilience to drought, salinity, and cold stresses, in addition to infection by rice blast disease which corresponded with their relative transcript abundance for OsSRDP. Altogether, our findings established that stress-inducible expression of *OsSRDP* can significantly enhance tolerance to multiple abiotic stresses and a biotic stress.

Front. Plant Sci., 13: 947312 (2023)

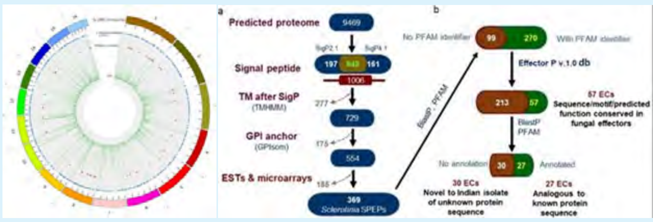


Functional characterization of a novel gene, DUF740 for multiple stress tolerance in rice

Draft genome sequencing and secretome profiling of *Sclerotinia sclerotiorum* revealed effector repertoire diversity and allied broad-host range necrotrophy

ICAR-NIPB in collaboration with ICAR-DRMR, Bharatpur, have decoded the draft genome sequence of an Indian isolate 'ESR-01' of the white mold (*S. sclerotiorum*) causing the stem rot disease in oilseed *Brassica*. The ~41 Mb genome of this pathogen harbours 9469 protein-coding genes out of which 529 genes were identified as carbohydrate-active enzymes (CAZymes) and 156 genes as essential for the pathogen-host interaction. The secretome profiling has revealed a total of 57 effector candidates (ECs) that facilitate disease development in the host plants.

Scientific Reports 12: 21855 (2022)

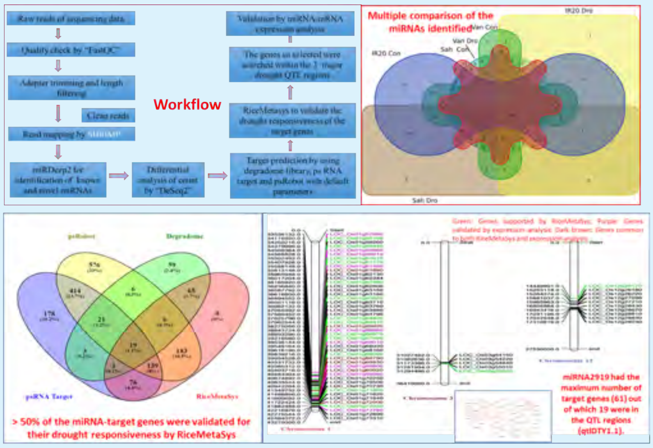


Whole genome sequencing and secretome profiling of *Sclerotinia sclerotiorum*

Integration of miRNA dynamics and drought tolerant QTLs (qDTYs) in rice reveals the role of miR2919 in drought stress response

To understand the role of miRNA in drought stress response and their relation, if any, with the qDTYs, the miRNA dynamics under drought stress was studied at booting stage in rice. Fifty-three known and 40 novel differentially expressed (DE) miRNAs were identified. Sixty-one target genes that corresponded to 11 known and 4 novel DE miRNAs were found to be co-localized with the three qDTYs. From our study, three known and novel miRNAs and seven target genes were identified as the most promising candidates for drought tolerance at booting stage. Of these, Osa-MIR2919 with 19 target genes in the qDTYs is being reported for the first time. It acts as a negative regulator of drought stress tolerance by modulating the cytokinin and brassinosteroid signaling pathway.

BMC genomics, 24: 526 (2023)

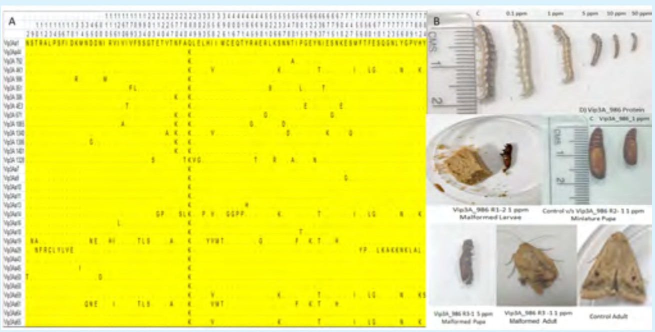


Identification of drought stress specific miRNA-mRNA modules in rice

Cloning, characterization and evaluation of toxicity of newly identified Vip3Aa proteins from *Bacillus thuringiensis* (*Bt*) for biological control of *Helicoverpa armigera*

Cloning and sequence characterization of 10 full-length *vip3*-type genes of the *Bt* isolates recovered from various Indian agro-climatic zones along with two known *Bt* strains revealed new amino acid substitutions which grouped them into six phylogenetic clusters. The immunostrip assays, SDS-PAGE and western blot analysis with Vip3A antibodies confirmed the presence of relevant Vip3A type proteins. The toxicity studies against *H. armigera* revealed LC_{50} of 0.921-8.513 ppm indicating their insecticidal potential. The Vip3A proteins were found to have toxic and sublethal effects on the growth and fecundity of the insect population. This study reveals the potential of the newly identified Vip3Aa proteins in the biological control of *H. armigera*.

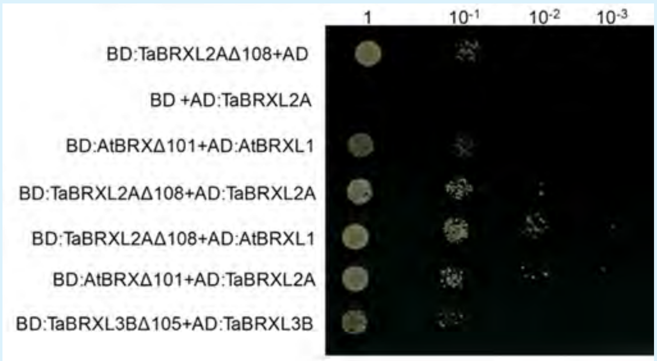
Journal of Pest Science. 1-24. doi.org/10.1007/s10340-023-01661-1



Allele mining for novel Vip3Aa proteins for biological control of *Helicoverpa armigera*

Genome wide analysis of BREVIS RADIX gene family from wheat (*Triticum aestivum*): A conserved gene family differentially regulated by hormones and abiotic stresses

BREVIS RADIX is a plant specific gene family with unique protein-protein interaction domain. We identified five BRX family genes in wheat and observed evolutionary conservation among BRX proteins from monocot species. Expression analyses showed abundance of TaBRXL1 transcripts in vegetative and reproductive tissues except flag leaf. TaBRXL5-A expressed exclusively in stamens. TaBRXL1 was upregulated under biotic stresses while TaBRXL2 expression was enhanced under abiotic stresses. TaBRX proteins exhibited homotypic (with BRX family form wheat) and heterotypic (with BRX family from Arabidopsis) interactions which corroborated with the role of BRX domain in protein-protein interaction. This study provided leads for functional characterization of TaBRX genes.

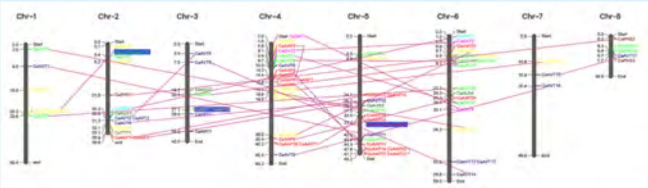


TaBRX protein is capable of homotypic, (within BRX family from wheat) and heterotypic (with BRX protein from *Arabidopsis*) protein - protein interaction.

Int. J Biol. Macromol. 232: 123081 (2023)

Genome wide identification and characterization of the amino acid transporter (AAT) genes regulating seed protein content in chickpea (*Cicer arietinum* L.).

Amino acid transporters (AATs), besides, being a crucial component for nutrient partitioning system are also vital for growth and development of the plants and stress resilience. A comprehensive analysis of AAT gene family in chickpea identified 109 AAT genes, representing 10 subfamilies, which were randomly distributed across the genome. Mature seed transcriptome data revealed that high protein content genotypes (ICC 8397, ICC 13461) showed low CaAATs expression as compared to the low protein content ones (FG 212, BG 3054). A significant difference in the amount of essential and non-essential amino acids, probably due to differential expression of CaAATs, was found in these genotypes. The insights gained from the present study will facilitate the functional characterization of AATs with respect to its diverse roles.

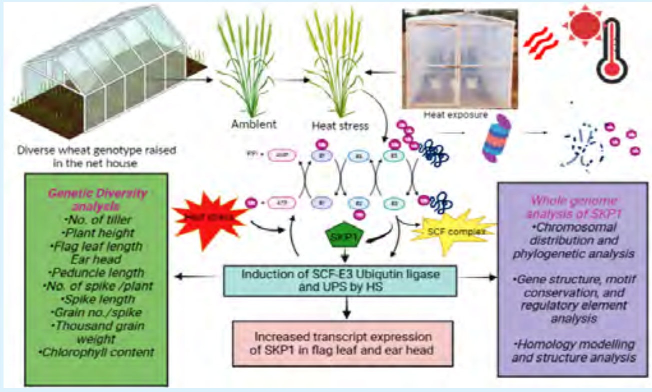


The distribution and duplication events of AAT genes across the chickpea genome

Int. J Biol. Macromol. 252: 126324 (2023)

Genetic diversity, transcript heterogeneity and allele mining of *TaSKP1-6B-4* gene variants across diverse genotypes under terminal heat stress and genome wide characterization of *TaSKP1* gene family from bread wheat (*Triticum aestivum* L.)

SKP1 (S-phase kinase protein1) is an essential regulatory component of a class of E3 ubiquitin ligases involved in maintenance of cellular protein homeostasis through ubiquitin mediated proteasome system (UPS). To assess the functional role of SKP1, which was earlier isolated at ICAR-NIPB, allele mining and transcript profiling was carried out in 25 contrasting germplasm of wheat grown under terminal heats stress (THS) in field. The cv. HD 2967 had the maximum upregulation under THS. Genome wide search of SKP1 gene family identified 95 SKP1 genes which were structurally characterized. Selected genotypes can be utilized in crop improvement and breeding programs for development of wheat varieties exhibiting heat tolerance at the terminal growth stage through conventional as well as new age technologies.



Schematic representation of the role of SKP1 in wheat for terminal heat stress tolerance

Plant Mol Biol. PMID: 37985582 (2023)

EVENTS

ICAR-NIPB Co-hosted an International Conference

ICAR-NIPB co-organized an International Conference on Food and Nutritional Security (iFANS-2023) at National Agri-Food Biotechnology Institute, Mohali on 6th - 9th January 2023. More than 450 delegates from India and other countries participated in the conference. The conference was inaugurated by Hon'ble Minister of State (IC) Dr. Jitendra Singh, Ministry of S&T.

Inauguration of iFANS-2023

Foundation Day Celebration

ICAR-NIPB celebrated Foundation Day on 15th January 2023. Dr. Rajesh S. Gokhale, Secretary, Department of Biotechnology, GoI, delivered Foundation Day lecture on "Fostering Biotech Innovation to Accelerate Bioeconomy." Dr. T. Mohapatra (Former Secretary DARE and DG, ICAR) presided over the function. Many dignitaries, including Dr. T. R. Sharma (DDG, Crop Science) and Dr. D. K. Yadava (ADG, Seeds) attended the function.

ICAR-NIPB foundation day celebration

Farewell to Dr. Ajit Shasany, Director, ICAR-NIPB

Upon selection as the Director of CSIR-National Botanical Research Institute, Lucknow, NIPB staff bid a farewell to Dr. A. K. Shasany for his significant contribution as the Director, ICAR-NIPB. NIPB staff wishes him good luck with his new assignment.

Farewell to Dr. Ajit K. Shasany

International Women's Day 2023

Every year worldwide 8th March is celebrated as United Nation's International Women's Day. To mark the day, a lecture was organized at ICAR-NIPB on *"Deconstructing Gender"*. The speaker on the occasion was Prof. Bulbul Dhar James, Professor (Political Science) and Former Director, Sarojini Naidu Centre for Women Studies, Jamia Millia Islamia, New Delhi. The talk was chaired by Dr Anita Grover, Director, ICAR-NIPB and organized by the womens' cell, ICAR-NIPB.

International Women's day celebration

Review Meeting of ICAR-NPTC/FGGM/TGCP

The 15th review meeting of ICAR-NPTC/FGGM/TGCP was conducted on March 15 under the

Review meeting of ICAR-NPTC/FGGM/TGCP

chairmanship of Prof. Swapan Datta (Former DDG, CS). Translation genomics work in five crops- rice, wheat, pigeonpea, jute, and mango was reviewed by Dr D.K. Yadava, Dr Anita Grover, Prof. N.K. Singh and Prof. P.C. Sharma as expert committee members. Dr P.K. Dash was the convener.

Participated in International Yoga Day

ICAR-NIPB participated in the International Yoga Day on 21st June 2023 at NASC complex.

International Yoga Day celebrations

Dr Ramcharan Bhattacharya joined as the Director

Dr. Ramcharan Bhattacharya (Principal Scientist, NIPB) joined as the newly appointed Director of ICAR-NIPB on August 2, 2023. Previously, he served as the Group Leader of the Brassica Improvement project of NIPB.

Joining of Dr. Ramcharan Bhattacharya as Director ICAR-NIPB

Institute Management Committee Meeting

The Institute Management Committee (IMC) was conducted on September 12, under the chairmanship of Dr. R.C. Bhattacharya, Director, NIPB. Members of IMC and special invitees discussed several key issues of the institute. Members of the IMC present in the meeting were Dr. D.K. Yadava (ADG, Seed, ICAR), Dr. Rajender Parsad (Director, IASRI, New Delhi), Dr. S.K. Chadha (IFS, Special Secretary, Department of Agriculture & Technology, Government of Odisha), Dr. N.P. Singh (Vice-Chancellor, BUAT, Banda), Prof.

S.S. Pathani (Tanmay Kuteer), Dr. Ratan Tiwari (Pr. Scientist, IIWBR, Karnal), Dr. Anuradha Upadhyay (Pr. Scientist, NRC Grapes), Shri. Saurabh Muni (SF&AO, IARI) and Shri. Sumit Singh (SAO, NIPB & Member Secretary). The special invitees from NIPB present in the meeting were Prof. N.K. Singh, Dr. Sarvjeet Kaur, Dr. D. Pattanayak, Dr. P.K. Mandal, Dr. T.K. Mondal, Dr. P.K. Jain, Dr. K. Gaikwad, Dr. P. Dash, Shri Rahul Kumar, Smt. Sangeeta Jain, Shri Vipin Kumar.

Group photo with IMC committee members

Institute Research Council Meeting

ICAR-NIPB completed its Institute Research Council Meeting 2023 on September 18-19, 2023, under the chairmanship of Dr. R.C. Bhattacharya (Director, NIPB). The two-day presentation of research progress by all the scientists, ensuing discussions, critical remarks, and suggestions on each project led to retrospection and midterm corrections for achieving the targets. The two resource persons, Prof. Srinivasan (Former Director, NIPB) and Dr. MK Reddy (Arturo Falaschi Emeritus Scientist, ICGEB, New Delhi), gave important suggestions in their remarks. NIPB thank them for making many such critical suggestions. The Chairman (Dr. R.C. Bhattacharya) sensitized scientists to orient their research to fulfill the ICAR and the Nation's demands.

NIPB Scientists along with IRC experts

Awareness on International Year of Millets 2023

ICAR-NIPB organized a lecture on Millets for Food Security and Sustainability by Dr. Trilochan Mohapatra, Former Secretary, DARE and Ex-Director General, ICAR, on the occasion of the International Year of Millets 2023 as part of an initiative to raise awareness about millets.

NIPB family with Dr. T. Mohapatra

ICAR-NIPB Co-organized an International Conference

ICAR-NIPB in association with the Society for Plant Biochemistry and Biotechnology, New Delhi; Indian Agricultural Research Institute, New Delhi; and National Botanical Research Institute, Lucknow, organized an International Conference on Biochemistry and Biotechnological Approaches for Crop Improvement- 2023 at National Agricultural Science Complex, New Delhi from 30th October to 1st November, 2023. More than 450 delegates across the globe participated in the conference.

Inauguration of iBBAC 2023

Research Advisory Committee Meeting

The 14th Research Advisory Committee (RAC) meeting of the ICAR-NIPB, New Delhi was held on 19th December 2023 at NIPB Auditorium. The meeting was chaired by Prof. Sudhir Kumar Sopory, Emeritus Senior Scientist, International Centre for Genetic Engineering and Biotechnology (ICGEB) and former Vice Chancellor JNU, New Delhi. The esteemed members of RAC, along with Dr. R.C. Bhattacharya, Director ICAR-NIPB, and all scientific staff attended the meeting.

The NIPB family along with RAC committee experts felicitated Prof. Rajeev Varshney Director, Centre for Crop & Food Innovation, Murdoch University, Australia for his selection as Fellow of Royal Society (FRS) London.

Presentation by Director ICAR-NIPB in RAC meeting

Felicitation of Prof. Rajeev Varshney for FRS London

Swacchta Abhiyan from 16th-31st December 2023

ICAR-NIPB celebrated "Swachhata Pakhwada" during 16th - 31st December, 2023. Several activities including swachhata pledge, cleanliness and sanitation drive, campaign, awareness and workshop were organized daily.

Swachhata Abhiyan at ICAR-NIPB

Participation in ICAR Zonal Sports Tournament

A team of ICAR-NIPB staff comprising of scientific, technical and administration staff participated in Zonal Sports Tournament during 18th-21st December 2023 held at ICAR-CIAE, Bhopal. Ms. Nidhi Nailwal received Silver medal in the shot put event.

ICAR-NIPB Staff at Zonal Sports meet 2023

NEW PROJECTS SANCTIONED

Name	Project Title	Date of Initiation	Funding Agency
Dr. Joshitha Vijayan	Designing bidirectional promoter capable of driving Cas9 gene and guide RNA simultaneously for generating CRISPR-Cas9 based knockouts in soybean	11 July 2023	SERB
Dr. Rhitu Rai	Elucidating the underlying molecular mechanism for bacterial blight disease regulation by rice susceptibility gene, *OsTFX1*	15 Sept 2023	SERB Power
Dr. Mahesh Rao	Deciphering the genomic and cytogenetic factors for the polyploid evolution, speciation, intergenomic interactions and diversification of amphidiploid Brassica	11 Dec 2023	SERB-CRG

Publications

A total of one hundred and twenty nine publications including research articles in international peer reviewed high impact factor scientific journals, book and book chapters were published by ICAR-NIPB scientists during the reporting period. Also four promising germplasm were registered at PGRC ICAR-NBPGR, New Delhi.

Publication type	Number
Research articles	120
Books	2
Book chapters	7
Germplasm registration	4

Review and report of the loss at farmer's field due to heavy rain and the hailstorm

A team of ICAR-NIPB, Delhi comprising Dr. Rekha Kansal, Dr. R. C. Bhattacharya, Dr. Rohit Chamola and Dr. Mahesh Rao, visited different villages of district Bulandsahar, Uttar Pradesh on 21st March 2023 to meet the farmers and get an overview of the damage of crops due to heavy rain and the hailstorm during March 2023.

Farmer's field visit by ICAR-NIPB Staff

Krishi Vigyan Mela 2023

ICAR-NIPB, Delhi participated in Krishi Vigyan Mela 2023 from 2nd-4th March 2023 at ICAR-IARI, New Delhi where the research activities, achievements and technologies of the institute were displayed. The staff and students of ICAR-NIPB actively participated and many visitors including farmers, researchers and students visited the institute stall and got the information related to the institute activities. A demonstration of DNA isolation from banana was conducted at the stall which attracted the visitors especially students and agriculture graduates.

ICAR-NIPB stall at Pusa Krishi Vigyan Mela 2023

Scheduled Caste Sub Plan (SCSP) Program

ICAR-NIPB, Delhi organised the farmer's training and farmers kit distribution program under Scheduled Caste Sub Plan (SCSP) program of Govt. of India at different villages across the country. SCSP programmes were organised with the help of the local officials and KVKs and the staff of ICAR-NIPB Delhi actively participated and organised these programmes at different locations viz., at KVK Amihit, Jaunpur, Uttar Pradesh on 27th September 2023; Deopokhar, Kushinagar, U.P. on 7th October 2023; at Amwa, Deoria, U.P. on 8th October 2023; at Jaitpura, Phulera, Jaipur, Rajasthan on 25th October 2023; at Mankada, Malappuram, Kerala on 26th October 2023; at Uttaron, Uttarakhand on 3rd November 2023; and at Chargaon, Warora, Maharashtra on 22nd December 2023. A large number of farmers (mostly from SC/ST communities), including men and women farmers, attended these trainings and also the seed kits of vegetables, cereals, and plant saplings were distributed among the farmers.

SCSP program at KVK Amihit, Jaunpur, U.P.

SCSP program at Deopokhar, Kushinagar, U.P.

SCSP program at Amwa, Deoria, U.P.

SCSP program at Jaitpura, Phulera, Jaipur, Rajasthan

SCSP program at Mankada, Malappuram, Kerala

SCSP program at Uttaron, Uttarakhand

SCSP program at Chargaon, Warora, Maharashtra

OUTREACH: TRAINING AND VISITS

International Visitors

Two very interesting talks on "Molecular mechanism of auxin biosynthesis and degradation" and "Function and evolution of phased, secondary siRNAs (Phasi) in plant reproduction and other pathways" were delivered by Prof. Yunde Zhao and Prof. Blake Meyers, respectively on 11th January 2023 at NIPB.

Lecture by Prof. Yunde Zhao

Lecture by Prof. Blake Meyers

ICAR-NIPB Family along with Prof. Yunde Zhao and Prof. Blake Meyers

Prof. Motoyoki Ashikari from Nagoya University, Japan visited NIPB on 7th November 2023 and delivered a talk on “Molecular mechanism of stem elongation in rice".

ICAR-NIPB Family along with Prof. Motoyoki Ashikari

School/college students visit at NIPB

Outreach Program for Biological Sciences for School Children

To create awareness and interest in the young minds in the area of biology an outreach program

Outreach program for Carrier Opportunities in Biological Sciences at Modern Vidya Niketan Senior Secondary School, Sector-17, Faridabad was conducted by ICAR-NIPB along with Modern Vidya Niketan Senior Secondary School, Sector-17, Faridabad. Drs Pranab Kumar Mandal, Amitha Mithra and Amolkumar Solanke served as the resources persons. They gave a detailed presentation on the biological courses at college level, research and job opportunities available in the country and abroad.

Student Visit to ICAR-NIPB, Delhi

A large number of students from various universities and schools from all over India visited ICAR-NIPB during January to December, 2023. Around 156 B.Sc. Agriculture students from College of Agriculture, Vellayani, Thiruvananthapuram, Kerala along with faculty visited ICAR-NIPB on 10th March 2023. Around 64 B.Tech Biotechnology students from Shri. Shivaji college of Agricultural Biotechnology Amaravati visited ICAR-NIPB on 30th October 2023. Also students from Kerala Senior School, Vikaspuri, New Delhi; SGT University, Gurugram; Viaan International School Preet Vihar visited ICAR-NIPB on 4th January 2023, 28th November 2023 and 14th December 2023, respectively. They visited different facilities and also interacted with scientists. The students were told about various technologies developed by ICAR-NIPB and their applications.

Students from Shri. Shivaji College of Agricultural Biotechnology Amaravati

Students from College of Agriculture, Vellayani, Thiruvananthapuram, Kerala

Lab visit at ICAR-NIPB by students

Students from Kerala Senior School, Vikaspuri

Group photo of student visitors at ICAR-NIPB

Group photo of student visitors at ICAR-NIPB

Students form Viaan International School Preet Vihar

Students form SGT University Gurugram

Trainings

ICAR-NIPB arranges a long-term training programme in the area of Plant Molecular Biology and Biotechnology twice a year. The training aims at providing an opportunity for carry out research in the premier institutes with an exposure to all modern facilities. This is a paid training programme, where the students from the various universities are trained in the area from basic molecular biology to different advanced techniques like plant tissue culture and transformation, gene cloning, gene characterization, genome editing, sequencing, transcriptomics, genomics and proteomics, bioinformatics. The selected candidate gets a chance to work in one of the on-going projects of the Institute under the supervision of scientists of ICAR-NIPB. On completion of project work, the candidates are allowed to submit a thesis/report/ dissertation to his/her institution with the signature of the scientist of ICAR-NIPB as Co-Supervisor. During the year 2023 around 50 students received training from ICAR-NIPB from various universities of the country such as Graphics Era (Deemed to be University), Dehradun; Amity University, NOIDA; Kumaun University, Nainital; Ch. Charan Singh University, Meerut; Deen Dayal Upadhyay Gorakhpur University, Gorakhpur and so on.

Scientific Social Responsibility under the SERB-CRG project: A one-day program was held at ICAR-NIPB on 07th November 2023 in which 22 students from Kirorimal College, Delhi were given scientific insights, laboratory, and field exposure.

Group photo of students from Kirorimal College, Delhi

Inauguration of One Day Workshop

Participants of Workshop

One-day workshop on “Utilization of Plant Genetic Resources and Application of Genomic Tools for Efficient Pre-breeding Research in Crops” has been organized at ICAR-NIPB on 4th Oct, 2023.

MoUs /MTAs signed

A research MOU has been established between NIPB and King Abdulla University of Science and Technology, Saudi Arabia to work on “population genetics and *de novo* domestication of *Oryza coaractata*”

A MTA was signed with the Indian Institutes of Science Education and Research (IISER), Thiruvananthapuram, Kerala for sharing of genetic stocks of rapeseed-mustard for research purposes.

Dr Mahesh Rao, Senior Scientist visited The University of Bonn, Germany during 1st September

VISITS AND EXCHANGES

2022 to 28th February 2023 (six months) for training on “Breaking the blocks: eliminating linkage drag in introgression breeding in rapeseed mustard” under SERB-SIRE program of DST, Government of India.

Dr. Anshul Watts, Scientist (SS), visited University of Nantes France under Dr. Bernard Offmann to get an overview of germination bioassay of parasitic plant *Orobanchae* from 28th January 2023 to 10th February 2023.

Visit of Dr Mahesh Rao at University of Bonn, Germany

Dr. Navin Chandra Gupta visited Australia to attend 16th International Rapeseed Congress (IRC-2023), from 24th-27th September 2023, at Sydney, Australia.

Dr. Ramcharan Bhattacharya, Director ICAR-NIPB, Delhi attended High level policy dialogue on Gene editing for sustainable agriculture and food security organised by APAARI and Asia Pacific Consortium on Agriculture Biotechnology and Bioresources at Bangkok Thailand during 11-12 December2023.

Dr. Ramcharan Bhattacharya in Bangkok, Thailand

Dr. N. C. Gupta at 16th IRC, Sydney, Australia

Visit of Dr. Anshul Watts at University of Nantes France

AWARDS AND HONOURS

Dr T.K. Mondal Received "Prof Gadgil Memorial Lecture" Awards 2024 by Plant Tissue Culture Association (India).

Dr P.K. Mandal attended IMC Meeting of as ICAR-SBI, Coimbatore as IMC member on 28th June 2023.

Dr Mahesh Rao delivered a presentation on "Host-Pathogen Interaction of *Alternaria brassicae* and *Brassica juncea* Introgression Lines" in the International conference BioSCon'23 "Advancement in Plant Health Research- Retrospect & Prospect" which was held on 7th-8th December 2023 at Visva-Bharati University, Santiniketan, West Bengal, India. He has also received "Young Scientist Award" in the Conference.

Dr Mahesh Rao, receiving Young Scientist Award

Dr T.K. Mondal delivered an invited talk on 23rd November 2023 on "Discovery and validation of novel QTL/gene associated with awn length variation in biparental population derived from an interspecific cross between *Oryza sativa* L. and *Oryza glaberrima* Steud" at International conference on "Next-Gen preparedness for food security and environmental sustainability" held during 21st to 24th November 2023 at Assam Agricultural University, Jorhat.

Dr. T.K. Mondal delivered an invited talk on Functional characterization of salt tolerant genes from *Oryza coarctata*: an triploid wild species of rice at International Symposium on Rice Functional genomics-2023, Bangalore held on 2nd to 5th November, 2023 at UAS, Bangalore.

Dr Subodh K Sinha delivered an invited talk on "Functional validation of a high-affinity nitrate transport system of bread wheat (*Triticum aestivum* L.)" in the International Conference on Biochemical and Biotechnological Approaches for Crop Improvement (IBBACI) 2023, at New Delhi, on 1st November, 2023.

Dr Navin Gupta delivered a invited talk on "Molecular profiling of the *Sclerotinia sclerotiorum* - Brassica pathosystem for controlling stem rot disease in mustard" in the 16th International Rapeseed Congress (IRC-2023) from 24th-27th September 2023, Sydney, Australia.

Dr. Navin C. Gupta delivered a presentation in International conference on vegetable oils 2023 at ICAR-IIOR, Hyderabad during 17th-21th January 2023 on "Secondary metabolite analysis in the broad host range in the plant -pathogenic fungus *S. sclerotiorum*" and received Best oral presentation Award.

Dr Amolkumar Solanke served as the Chief Guest at Botany Society – BLOOM, Ramjas College, Delhi University on 20th October 2023.

Dr Amolkumar Solanke delivered the following invited talks:

'Efforts Towards Development of Blast Resistance in Rice' at Botany Society – BLOOM, Ramjas College, Delhi University on 20th October 2023;

'Structural and functional genomics of tomato' during NAHEP-CAAST Training on 'Learning Genomics Tools and Techniques for Improvement of Vegetable Crops' at Division of Vegetable Sciences, Indian Agricultural Research Institute, New Delhi on 13th November 2023;

'Genomics approaches for controlling fungal plant pathogens' during NAHEP-CAAST Training on 'Population Diversity, Pathogenomics and Development of Diagnostics of Emerging Fungal Plant Pathogens' at Division of Plant Pathology, Indian Agricultural Research Institute, New Delhi on 28th November 2023;

'Whole Genome Sequencing Platforms: Strategies and Protocols' during Short term Training Programme on "Fungal Genome Sequencing: Basic Biology to Biotechnology" Division of Plant Pathology, Indian Agricultural Research Institute, New Delhi on 13th March 2023.

हिन्दी गतिविधियां

2023–हिंदी कार्यशाला

क्र.सं.	दिनांक	तिमाही	मुख्य व्याख्याता
1.	13.03.2023	जनवरी–मार्च	डॉ. श्री प्रेमपाल शर्मा, पूर्व सचिव, रेल मंत्रालय विषयः–"नई शिक्षानीति और भारतीय भाषाएँ"
2.	23.06.2023	अप्रैल–जून	डॉ. सुबोध कुमार सिन्हा, प्रधान वैज्ञानिक, राष्ट्रीय पादप जैवप्रौद्योगिकी संस्थान, नई दिल्ली विषयः– "राजभाषा कार्यान्वयनः संस्थानिक दायित्व एवं गतिविधियां"
3.	14.09.2023	जुलाई– सितंबर	डॉ. रामचरण भट्टाचार्या, निदेशक, भा.कृ.अ.प.–राष्ट्रीय पादप जैवप्रौधोयोगिकी संस्थान, नई दिल्ली विषयः– "दैनिक कामकाज में हिंदी भाषा का सरल प्रयोग"
4.	04.10.2023	अक्टूबर– दिसंबर	डॉ. डी. के. यादव, सहायक महानिदेशक (बीज), भारतीय कृषि अनुसंधान परिषद, कृषि भवन, नई दिल्ली विषयः–"हिंदी के प्रति अपने दायित्व"

हिंदी कार्यान्वयन समिति की बैठक

1. दिनांक17–02–2023 (प्रथम तिमाही, जनवरी से मार्च, 2023)
2. दिनांक 08–06–2023 (द्वितिय तिमाही, अप्रैल से जून, 2023)
3. दिनांक 17–08–2023 (तृतीय तिमाही, जुलाई से सितंबर, 2023)
4. दिनांक 26–10–2023 (चतुर्थ तिमाही, अक्टूबर से दिसम्बर, 2023)

संस्थान में हिंदी कार्यक्रम

हिंदी चेतना मास में निबन्ध प्रतियोगिता

NIPBians' Cricket Extravaganza: A Perfect Blend of Science and Sports

ICAR-NIPB organized a cricket match on 4^{th} February 2023, where scientists, administrative staff, MBB students, research associates, SRFs and other staff came together.

MBB students during cricket match

Farewell to degree recipients and Freshers' welcome for newly joined MBB students

During 61^{st} Convocation 2023, six Ph.D. and nine M.Sc. students earned their degrees. This year 15 Ph.D., 7 M.Sc. and 14 BTech Biotech students joined Molecular Biology and Biotechnology discipline. A farewell programme for the degree recipients as well as freshers programme for the new students were organized.

MBB students along with NIPB faculty

First batch of B.Tech. Biotechnology started at MBB-NIPB

It is a historic moment for ICAR-NIPB to welcome the first batch of undergraduate students for B.Tech. Degree in Biotechnology. A total 25 B.Tech. students joined the institute in the year 2022-23.

Fresher's welcome programme

International Exposure

The following students from Molecular Biology and Biotechnology Division of this Institute visited various international institutes to carryout part of their Ph.D. work under ICAR-National Agricultural Higher Education Project- Centre for Advanced Agricultural Science and Technology (NAHEP-CAAST) fellowship.

- Maryam Mukhtar visited The Sainsbury Laboratory, Norwich United Kingdom from 18^{th} January to 20^{th} March 2023.

- Priya Upadhyay visited INRES, University of Bonn, Germany from 1^{st} February to 21^{st} March 2023.

- Gopal Kalwan visited Murdoch University, Perth, Western Australia from 24^{th} October to 15^{th} December 2023.

- Jeet Roy visited IRRI, Manila, Philippines from 22^{nd} October to 16^{th} December 2023.

- Meena S. visited Utah State University, Logan, USA from 29^{th} November to 18^{th} December 2023.

- Nitasana Rajkumari visited National Chung Hsing University, Taichung, Taiwan from 18^{th} to 28^{th} December 2023.

PERSONNELIA

New Assignments

- Dr. Ramcharan Bhattacharya, Principal Scientist of this Institute joined as Director, ICAR-NIPB and resumed the charge on 2nd August, 2023.

- Dr. Pawan Kumar Agrawal, Principal Scientist of this Institute joined as Joint Director Education at ICAR-National Institute of Biotic Stress Management, Raipur on 4th August 2023.

- Dr. Ajit Kumar Shasany, Director, ICAR-NIPB has joined in a new assignment as Director, CSIR-NBRI, Lucknow on 2nd February 2023.

New Joining

- Smt. Sangeeta Malhotra, Private Secretary has joined ICAR-NIPB w.e.f. 14th November, 2023 after transferred from ICAR-NCIPM, New Delhi.

Transfer

- Sh. Anshul Kumar Verma, Technical Assistant transferred from ICAR-NIPB, New Delhi to ICAR-NBFGR, Lucknow on 2nd June 2023.

Retirements

- Dr. Anita Grover, Principal Scientist superannuated from ICAR services w.e.f. 30th April, 2023

- Dr. Rekha Kansal, Principal Scientist superannuated from ICAR services w.e.f. 30th April, 2023

Dr. Rohit Chamola, Chief Technical Officer was superannuated from ICAR services w.e.f. 30th September, 2023.

- Dr. N.K. Singh, National Professor has completed his tenure on 14th October, 2023.

FORTHCOMING EVENTS

- New Year's Day Celebration
- ICAR-NIPB Foundation Day 2024
- Felicitation of Degree Recipients
- CAFT training on, "Genome utilization and editing of plant for useful trait" from 7th - 27th February 2024.
- DST-SERB sponsored Hands-on training programme on Pre-breeding and cytogenetic approaches for crop improvement from 16th - 25th January 2024.
- Participation in Krishi Vigyan Mela, 2024
- International Yoga Day Celebration

Credit Lines:

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